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# PALEOMAGNETISM OF TIRUPATI SANDSTONES FROM GODAVARY VALLEY, INDIA

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The paleomagnetism of 68 samples (258 discs) of red sandstones belonging to the Tirupati formation of Middle Cretaceous age from Godavary Valley has been studied. The samples were collected from four different sites in an area of 10 square miles near Eluru (16°43'N, 81°10'E). The specimens were magnetically cleaned in alternating fields up to 800 oersted peak fields. The samples from three sites showed reverse polarization with down dips while at one site (No. 2) the samples had normal as well as reverse polarity, indicating the possibility of a geomagnetic field reversal during the period of magnetization of rocks. Mean declination and inclination for reversely magnetized samples from all four sites were N153°E and +56 (down dip) respectively and N329°E and -59° (up) for normally magnetized rocks from site 2. The corresponding Virtual Geomagnetic Pole is situated at 28°N, 73°W. These results are not much different from those of Lower Deccan traps of Upper Cretaceous age studied from various localities in India and lend a strong support to the hypothesis of Continental Drift of India since Cretaceous times. The results also support geomagnetic field reversal during Middle to Upper Cretaceous times.

## 1. INTRODUCTION

The paleomagnetic studies of Indian rocks have so far been concentrated mostly on the Deccan traps [1-3] which extend in age from Upper Cretaceous to Miocene and of Rajmahal and Sylhet traps of Middle Jurassic age [1, 4]. The results of these studies have strongly supported the continental drift theory of the Indian landmass between Jurassic to Miocene times from the southern hemisphere to its present position in the northern hemisphere.

In addition to continental drift the results of paleomagnetic studies on Deccan traps have given a strong evidence in support of geomagnetic field reversal during Upper Cretaceous times. The Deccan traps have been classified on the basis of paleomagnetic results into Lower and Upper Deccan traps, the Lower ones being reversely magnetized and the upper ones being normal. The plane of distinction between the normal and the reversed rocks seems to lie somewhere near an elevation of about 2000 ft above mean sea level [5].

In order to prove any major phenomenon such as continental drift or polar wandering it is desirable to study as many rock formations as possible for an evidence in its support or otherwise. A great deal of evidence in support of continental drift for Europe, North America, South America, Africa and Australia has come from the study of red sandstones and shales. A few of the well known formations studied are Triassic red beds and Ecca red beds of Lower Permian age from Africa [6, 7], Permian red beds of South America [8] and chocolate shales of Lower Triassic age from Australia [9]. In order to fill the age gap for paleomagnetic studies of Indian rocks from Carboniferous to Cretaceous age studies of Gondwana formations of Godavary Valley have been undertaken in the Institute. In this paper results of paleomagnetic studies on Tirupati red sandstones from a locality near Eluru are discussed.

## 2. GEOLOGY OF THE AREA AND SAMPLING

A well known exposure of Gondwana rocks in south India is Wardha-Godavary Valley which extends in a general NW-SE direction from Rajahmundry (17°N, 81°48'E) in the east to near Chanda (20°N, 79°21'E) in the west over a distance of nearly 450 km. The Upper Gondwana rocks are found between Rajahmundry and Vijayawada resting unconformably on Chintalpudi

Table 1
Stratigraphic sequence of rocks in Godavary Valley near Eluru.

| Age | Formation |
|---|---|
| Tertiary | Rajahmundry sandstones<br>Deccan traps |
| Upper Gondwanas | Tirupati sandstones<br>Raghavapuram shales<br>Gollapalli sandstones |
| Lower Gondwanas | Dummapet sandstones<br>Chintalpudi or Kamthi sandstones |
| Precambrian | Cuddapahs<br>Gneisses |

sandstones of Upper Permian age. Geologic sequence of rocks in the area is given in table 1 [10].

The general strike of the formations in the area is NE-SW. Dips are very gentle (less than 5°) towards SE. The Tirupatis are fine to medium grained red sandstones quarried locally for road construction.

Regarding the age of the formations there is some uncertainty. The most reliable estimate of their age has been given by L. F. Spath [11] who has examined Ammonites from Coastal Gondwana formations of Budavada sandstones (equivalent of Gollapalli) and Raghavapuram shales and has assigned the age of Upper Neocomian (Lower Cretaceous) to the Budavada sandstone. The Tirupati sandstones would then become probably Aptian (Lower or Middle Cretaceous).

The average width of Tirupati formation is 6 to 10 km, maximum length in NE-SW direction is about 60 km. Best exposures of these rocks were found near the southwest end of the outcrop. Samples were collected from four different outcrops near the village Janampeta (16°46'N, 81° 08'E) site 1, Padavegi (16°48'N, 81°09'E) site 2, near the village Nayanpalli (16°50'N, 81°12'E) site 3 and north east of Padavegi (16°49'N, 81° 13'E) site 4. At each site fresh oriented samples were collected from quarries which generally extended 100-200 m in length and about 2 m in thickness. From each sample three to four discs (specimens) of 2.5 cm diameter and about 2 cm length were cored and studied using an astatic magnetometer of sensitivity $3 \times 10^{-7}$ oersted/cm deflection at 4 m light path.

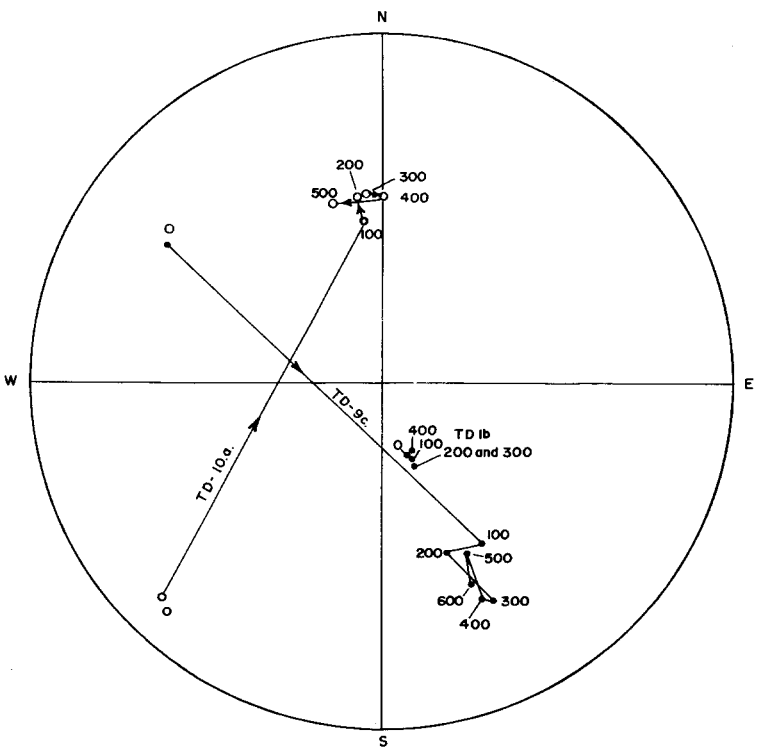



Fig. 1. Change in magnetization upon AC cleaning in various field strengths for 3 specimens from site 1 plotted on equal area projection. Figures denote peak alternating field strength in oersted. Down dip - ●; up dip - ○.

## 3. NRM MEASUREMENTS

From site 1 31 samples (103 specimens) were studied. The NRM measurements from this site showed mostly reverse polarization and down dips. The intensity of magnetization for most of the specimens lay between $1 \times 10^{-6}$ and $8 \times 10^{-6}$ emu/cm$^3$. The declination and inclination without AC cleaning were N130°E and +69° (down) respectively. From site 2, fifteen samples (65 specimens) were studied. These showed both normal as well as reverse polarization with up dip as well as down dip. The samples showed fairly wide scatter which was found later to be due to viscous magnetization. From site 3, 12 samples (51 specimens) were studied. Almost all samples (except one) showed reverse polarization with down dips. The mean of NRM directions was $D = 154°$E and $I = +48°$ (down). Mean intensity for all specimens was $4.5 \times 10^{-6}$ emu/cm$^3$. Scatter in NRM directions was found to be minimum for this site. From site 4 10 samples (39 specimens) were studied. All the specimens showed down dips with large scatter in declinations. The mean intensity of magnetization for all specimens was $1.40 \times 10^{-6}$ emu/cm$^3$.

## 4. AC DEMAGNETIZATION TESTS

A few specimens from various sites were subjected to alternating field demagnetization in progressively increasing peak fields of 50, 100, 200 oersted, etc. up to 800 oersted in an apparatus similar to that of Creer [12] and a certain value of the field was ascertained above which the direction of magnetization becomes stable. The rest of the specimens were treated at only two or three different values of the peak alternating field. The results for each site are discussed below.

The initial scatter in the directions of magnetizations which was found to be due to viscous magnetization, could be reduced considerably as a result of AC demagnetization in the case of all the sites. As a result of this treatment there was a considerable change in the direction of magnetization of some specimens towards their respective mean positions as is shown in fig. 1 for three specimens from site 1 and in fig. 2 for four specimens from site 2. Typical AC demagnetization curves of normalized intensity for few specimens from sites 1 and 2 (both normal and reverse) are shown in fig. 3.

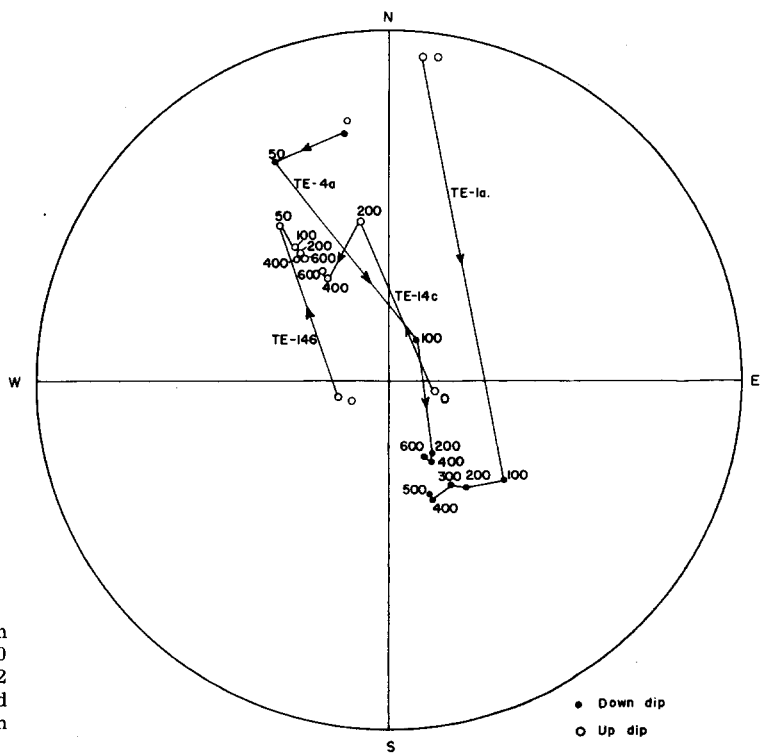



Fig. 2. Change in magnetization direction upon progressive demagnetizing up to 600 oersted field strength for 4 specimens (2 normal and 2 reverse) from site 2 plotted on equal area projection. Symbols as in fig. 1.

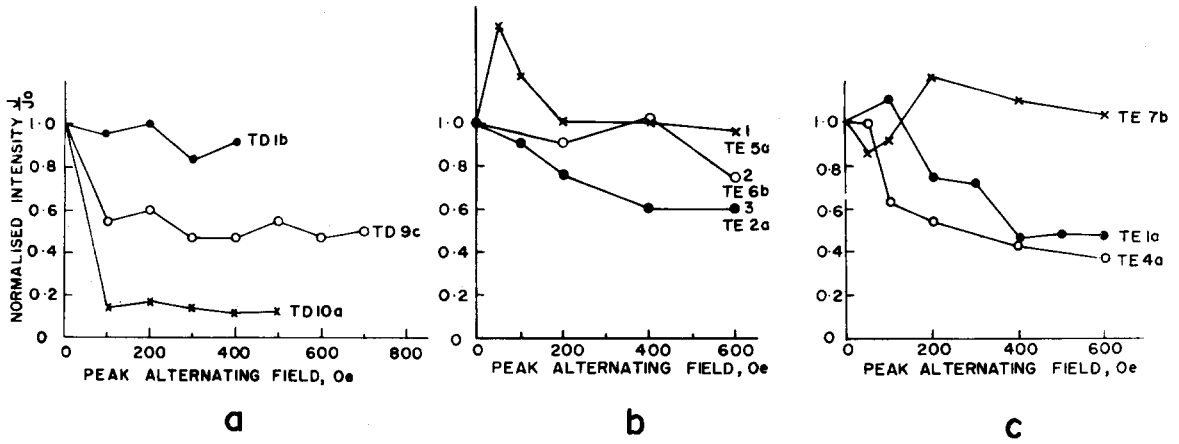



Fig. 3. (a) Typical AC demagnetization curves of normalized intensity for 3 specimens from site 1, (b) similar curves for 3 normal specimens from site 2, and (c) curves for 3 reverse specimens from site 2.

Most of the specimens from site 1 were demagnetized in 200 and 400 oersted peak fields and a few in a peak field of 800 oersted. Two specimens from this site were found to possess strong components of secondary magnetization which could not be cleaned adequately. These were excluded in computing the mean direction for this site. Specimens from site 2 were treated in a similar manner in peak alternating fields up to 600 oersted and those from site 3 and 4 were treated in peak fields up to 400 and 800 oersted respectively. The specimens of one sample from site 3 and eight specimens from site 4 were found to possess strong components of secondary magnetization which could not be cleaned even by treating at 800 oersted. These specimens were rejected in calculations of mean directions for these sites. Magnetically cleaned directions for each site are plotted on equal projection in figs. 4, 5, 6 and 7. These studies indicated that the samples from each site possessed stable directions of magnetization which were revealed as a result of AC cleaning. Samples from sites 1, 3 and 4 showed mostly reverse polarization with down dips while those from site 2 showed normal as well as reverse polarization almost 180°

Table 2
Paleomagnetic parameters for sites 1, 2, 3 and 4.

| Site no. | $N$ | $N_0$ | $n$ | $D$ | $I$ | $R$ | $k$ | $\alpha_{95}$ | $D$ | $I$ | $R$ | $k$ | $\alpha_{95}$ | VGP $\lambda_p^\circ$N | VGP $L_p^\circ$W | $\psi^\circ$ | $\lambda_m^\circ$S | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | After cleaning in 200 oersted | | | | | After cleaning in 600 oersted | | | | | | | | | |
| 1 | 31 | 2 | 94 | 157 | +51 | 28.65 | 80 | 3°14′ | | | | | | 36 | 75 | -23 | 28° | Reverse |
| 2 | 9 | - | 38 | 331 | -52 | 8.89 | 72.5 | 5°30′ | 329 | -59 | 8.97 | 267 | 3° | 26 | 73 | -31 | 35° | Normal |
| 2 | 6 | - | 26 | 156 | +61 | 5.92 | 62.5 | 7° | 154 | +62 | 5.97 | 167 | 4°30′ | 26 | 78 | -26 | 38° | Reverse |
| | | | | | | | | | After cleaning in 400 oersted | | | | | | | | | |
| 3 | 12 | 1 | 47 | 153 | +56 | 10.91 | 111 | 4° | 153 | +56 | 10.95 | 200 | 3° | 30 | 74 | -27 | 33° | Reverse |
| | | | | | | | | | After cleaning in 800 oersted | | | | | | | | | |
| 4 | 10 | - | 31 | | | | | | 145 | +55 | 9.97 | 300 | 2°38′ | 27 | 70 | -35 | 32° | Reverse |
| | Mean for all sites | | | | | | | | | | | | | | | | | |
| | 68 | 3 | 236 | | | | | | 153 | +56 | | | 4° | 28 | 73 | -29 | 34° | |

$N$ - total number of samples studied; $N_0$ - number of samples rejected for calculation of mean position; $n$ - number of specimens included for calculations; $D$ - declination of stable RM in degrees of E of N; $I$ - inclination of stable RM in degrees down +ve and up -ve; $R$ - length of vector sum of all individual vectors; $k$ - precision parameter; $\alpha_{95}$ - radius of 95% confidence circle; VGP - Virtual Geomagnetic Pole position; $\lambda_p$ - latitude, $L_p$ - longitude; $\psi$ - rotation in degrees; $\lambda_m$ - ancient latitude of Nagpur (centrally located place in India).

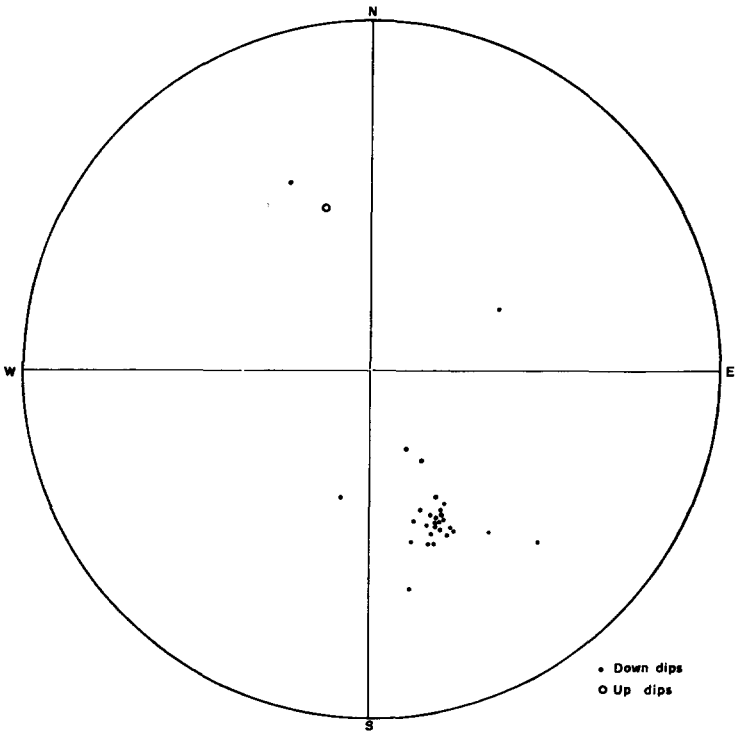



Fig. 4. Magnetically cleaned directions of magnetization for site 1 plotted on equal area projection. Symbols as in fig. 1.

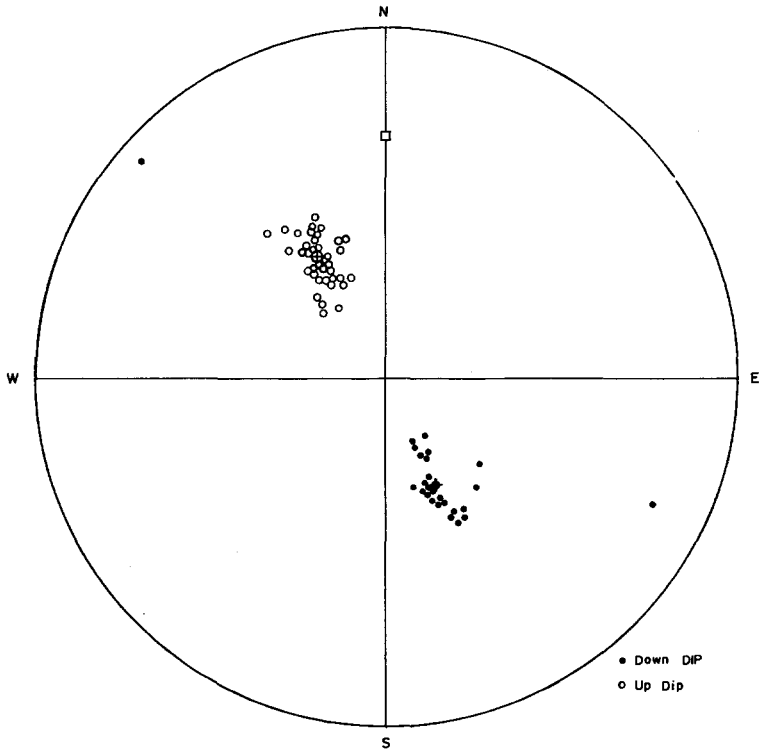



Fig. 5. Directions of magnetizations of normal and reverse specimens from site 2 after cleaning in 600 oersted peak fields. Mean direction for normal - ⊕; mean direction for reverse ●; present dipole field - □. Other symbols as in fig. 1.

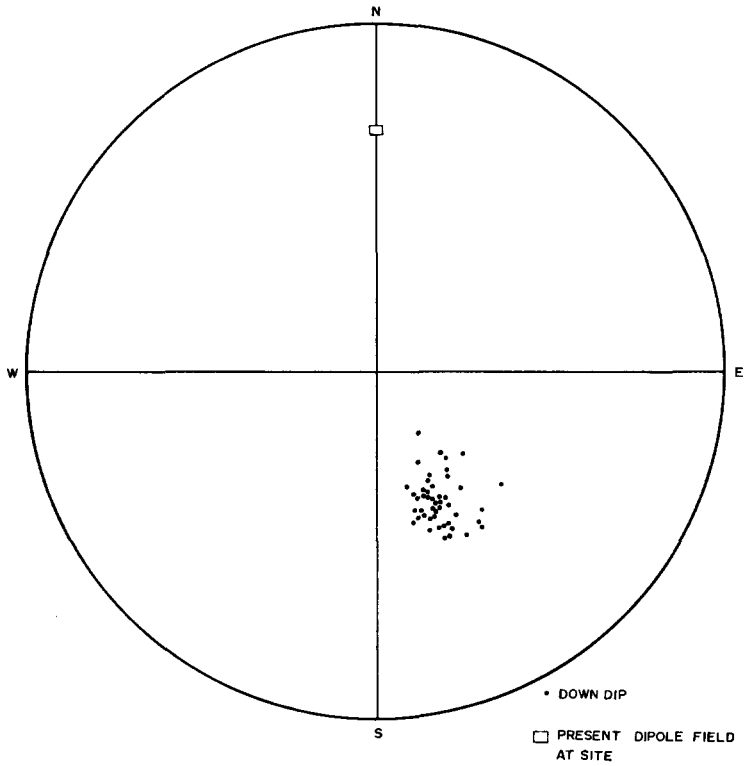



Fig. 6. Directions of magnetization of specimens from site 3 after magnetic cleaning in 400 oersted peak fields plotted on equal area projection.

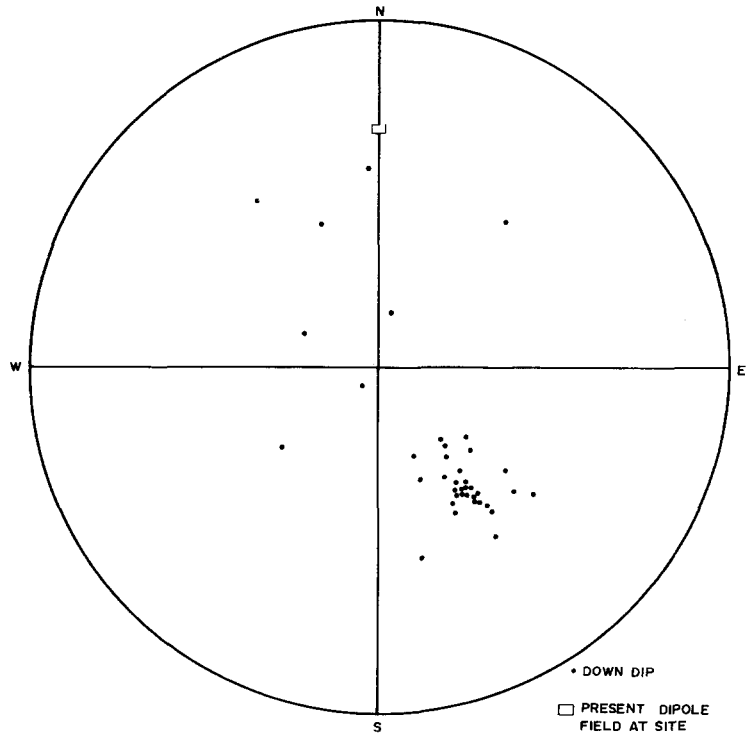



Fig. 7. Magnetic directions of specimens from site 4 after AC cleaning in 800 oersted peak fields.

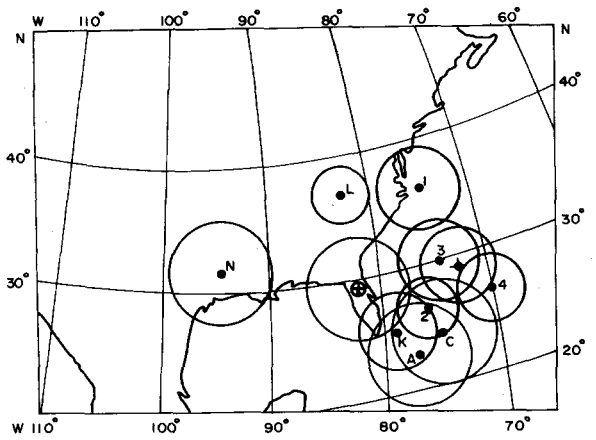



Fig. 8. Pole positions for Lower Deccan traps from various sites, Khandala - K, Linga - L, Amba - A, Nipani - N, Chinchli - C, compared with pole positions obtained on Tirupati sandstones from sites 1-4. ◆ mean pole position for Tirupati sandstones. Circles are drawn at 95% confidence level.

apart. The mean directions for each site and other paleomagnetic parameters are given in table 2.

## 5. RESULTS AND DISCUSSION

The results from the four sites can be best presented in terms of virtual geomagnetic pole (VGP) positions assuming a geocentric axial dipole field. For all sites magnetically cleaned directions were used for calculation of mean direction. These are given in table 2 along with other parameters. For calculation of VGP each sample direction was given a unit weight. For site 2 normal directions were reversed for pole position calculation. There is a fairly good agreement in pole position for each site.

It is of interest to compare these pole positions with those obtained from Lower Deccan traps (of Upper Cretaceous age) from various localities such as Amba, Nipani, Khandala, Linga [5] and Chincholi, etc. [3]. The pole positions are plotted in fig. 8.

It is seen that the pole position for the Tirupati sandstones is not significantly different from those of the Lower Deccan traps. The results therefore support the earlier paleomagnetic results from the Deccan traps and give strong evidence in support of continental drift of the Indian landmass from Lower Cretaceous to Miocene times.

It is not yet known whether the reverse rocks at site 2 represent a reversal of geomagnetic field or these are magnetized due to some self-reversing mechanism. However, the results give an indication of a possible geomagnetic field reversal during Lower Cretaceous times.

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